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Calibration results
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Normalized Residuals
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Reprojection error (cam0):     mean 0.11627410795011918, median 0.09741920390115585, std: 0.0789756452859909
Reprojection error (cam1):     mean 0.11423321615969884, median 0.09497176022790192, std: 0.07873384118174584
Gyroscope error (imu0):        mean 0.1288377321562226, median 0.12346572120522142, std: 0.0565436981578199
Accelerometer error (imu0):    mean 0.04631081987905688, median 0.03996703576921435, std: 0.026758662395424883

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 0.11627410795011918, median 0.09741920390115585, std:
0.0789756452859909
Reprojection error (cam1) [px]:     mean 0.11423321615969884, median 0.09497176022790192, std:
0.07873384118174585
Gyroscope error (imu0) [rad/s]:     mean 0.0027933991435149545, median 0.00267692572739425, std:
0.0012259538829331499
Accelerometer error (imu0) [m/s^2]: mean 0.011971406474509472, median 0.0103315301267404, std:
0.006917148629335091

Transformation (cam0):
------------------------
T_ci:  (imu0 to cam0):
[[ 0.99999678 -0.00088113  0.00238129  0.00350781]
 [ 0.00087383  0.99999492  0.00306542 -0.00110474]
 [-0.00238397 -0.00306333  0.99999247 -0.03589146]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0):
[[ 0.99999678  0.00087383 -0.00238397 -0.0035924 ]
 [-0.00088113  0.99999492 -0.00306333  0.00099788]
 [ 0.00238129  0.00306542  0.99999247  0.03588623]
 [ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.005462534568150308


Transformation (cam1):
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T_ci:  (imu0 to cam1):
[[ 0.99999808 -0.00054934  0.00188032 -0.04651982]
 [ 0.00054765  0.99999945  0.0008983  -0.00103816]
 [-0.00188082 -0.00089727  0.99999783 -0.03566075]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[ 0.99999808  0.00054765 -0.00188082  0.04645323]
 [-0.00054934  0.99999945 -0.00089727  0.00098061]
 [ 0.00188032  0.0008983   0.99999783  0.03574908]
 [ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.005392163953234804

Baselines:
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Baseline (cam0 to cam1):
[[ 0.99999982  0.00033026 -0.00050198 -0.05004528]
 [-0.00033135  0.9999976  -0.00216635 -0.00001002]
 [ 0.00050126  0.00216651  0.99999753  0.00023126]
 [ 0.          0.          0.          1.        ]]
baseline norm:  0.050045814036265006 [m]


Gravity vector in target coords: [m/s^2]
[-0.14703513 -9.80544674 -0.00422247]


Calibration configuration
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cam0
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```

Camera model: pinhole
Focal length: [387.26496788634023, 388.0178836701606]
Principal point: [320.53538193017073, 238.7961117373796]
Distortion model: radtan
Distortion coefficients: [-0.0048964025986568405, -0.0032256738628326904, 6.867803461190135e-05, 0.0003187277311499132]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

cam1

Camera model: pinhole
Focal length: [388.18861658777746, 388.9712538925426]
Principal point: [321.2136971603922, 238.95658293993972]
Distortion model: radtan
Distortion coefficients: [-0.002870215094047284, -0.00471202846643744, 5.440321557997829e-05, -9.458602566352592e-05]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

IMU configuration
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IMU0:

Model: calibrated
Update rate: 200.0

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Accelerometer:
Noise density: 0.01827880119715682
Noise density (discrete): 0.2585012855694074
Random walk: 0.0005154477377534344
Gyroscope:
Noise density: 0.0015331156827140093
Noise density (discrete): 0.02168152991181039
Random walk: 1.6734607279923772e-05
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
```
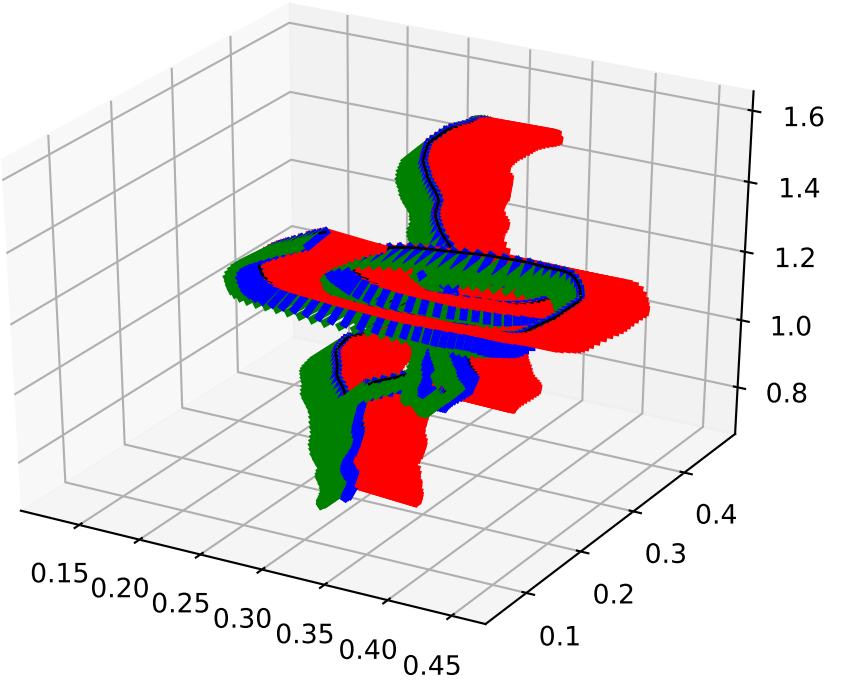
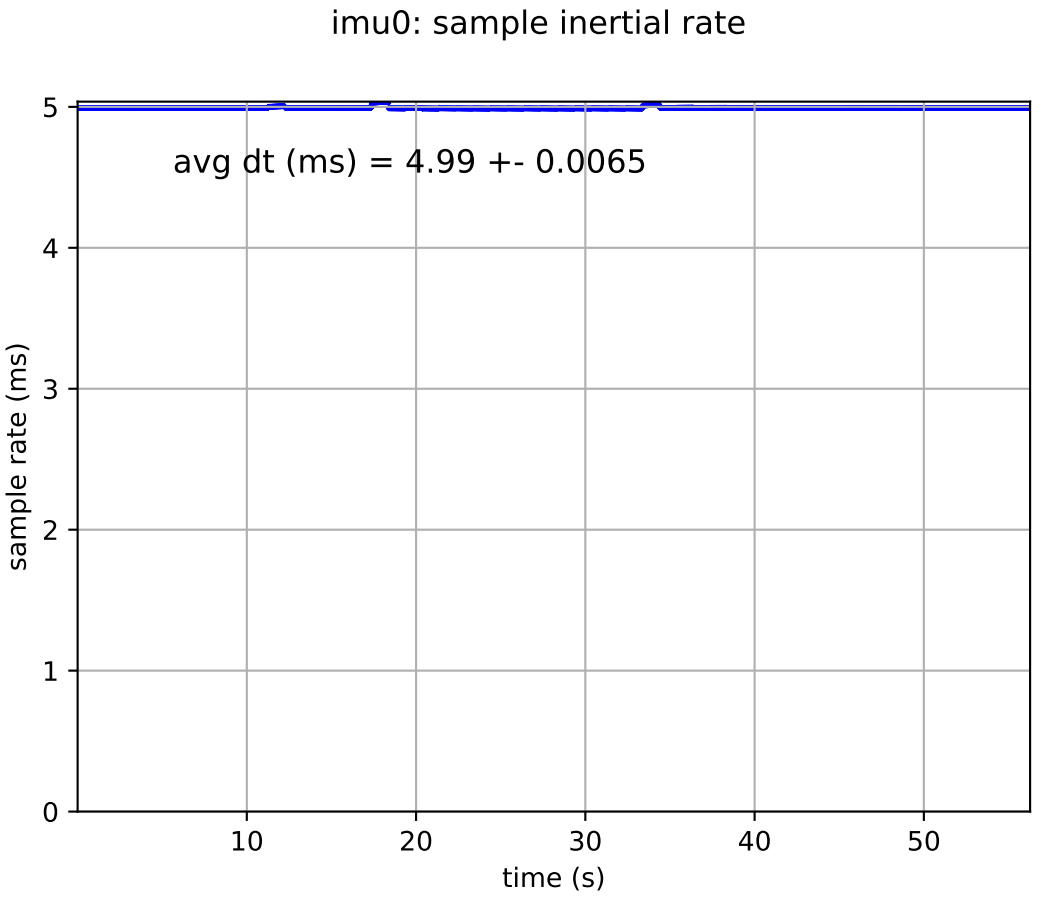
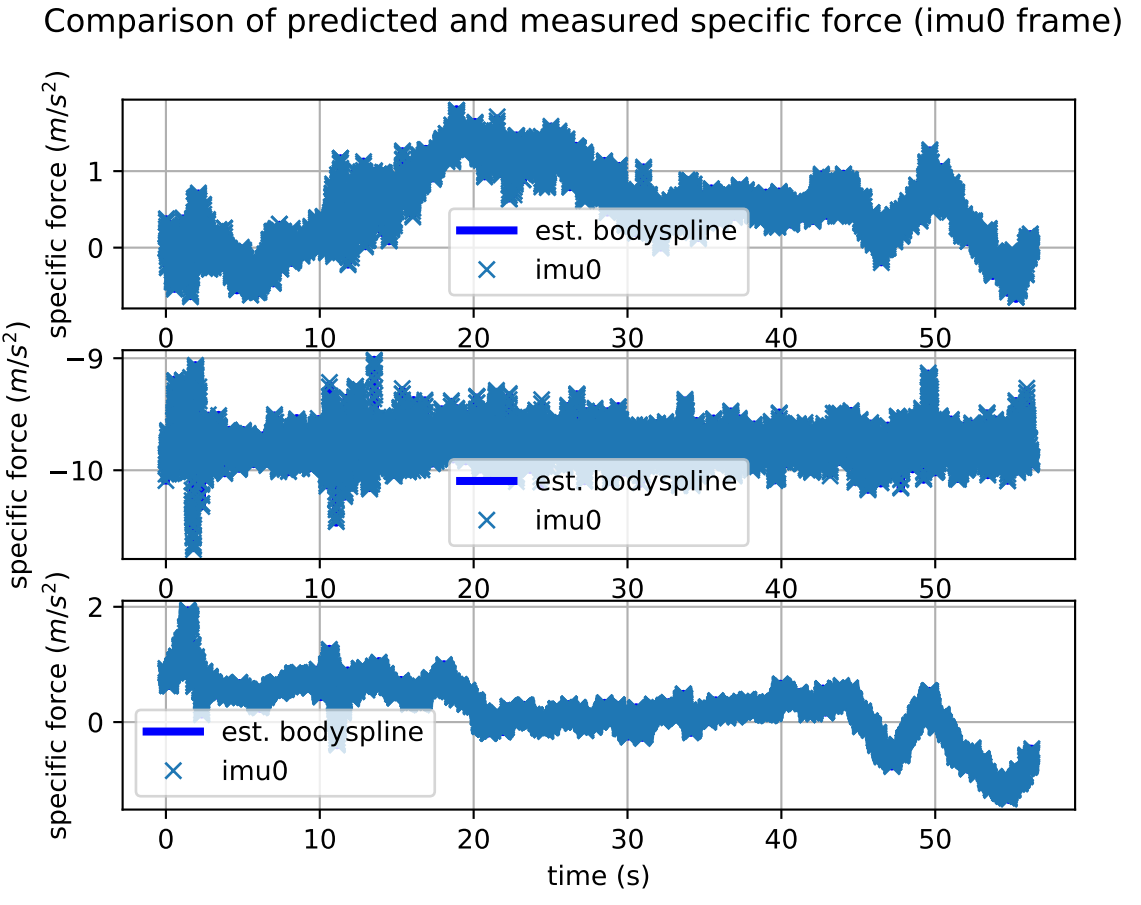
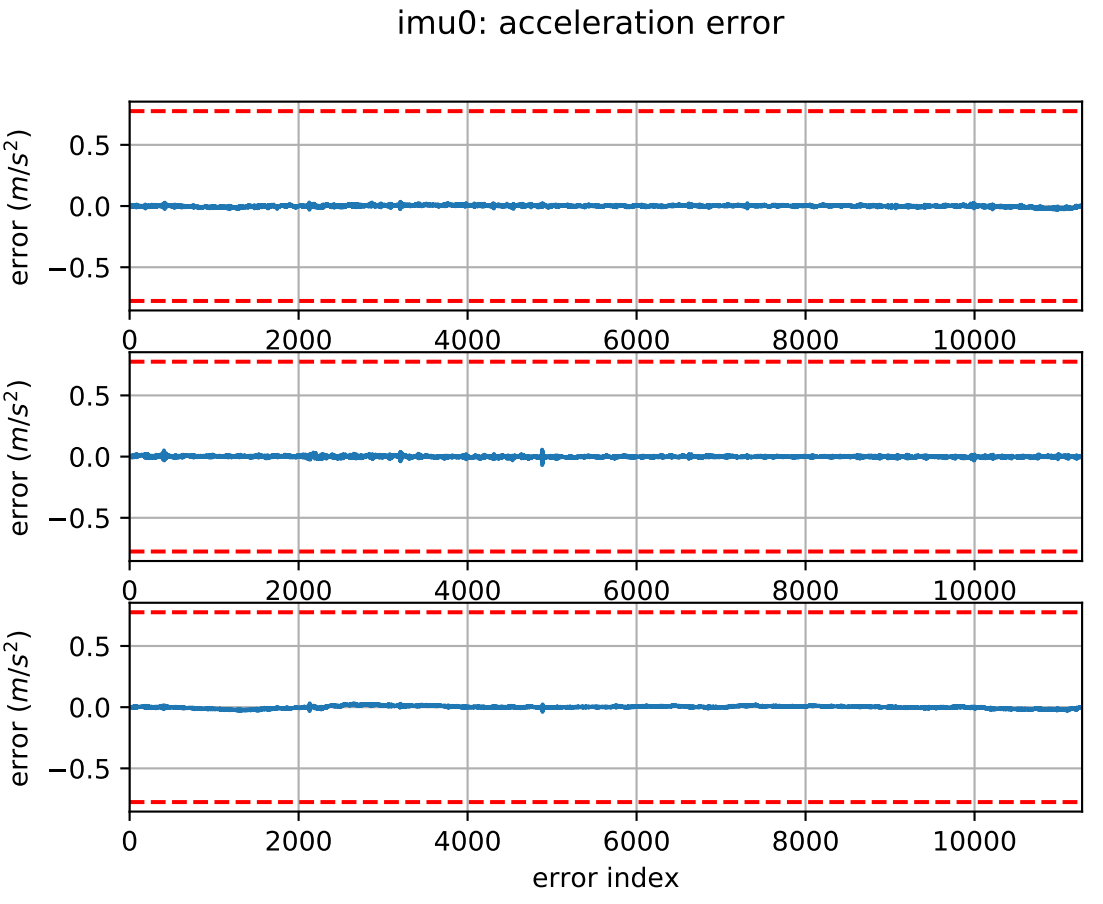
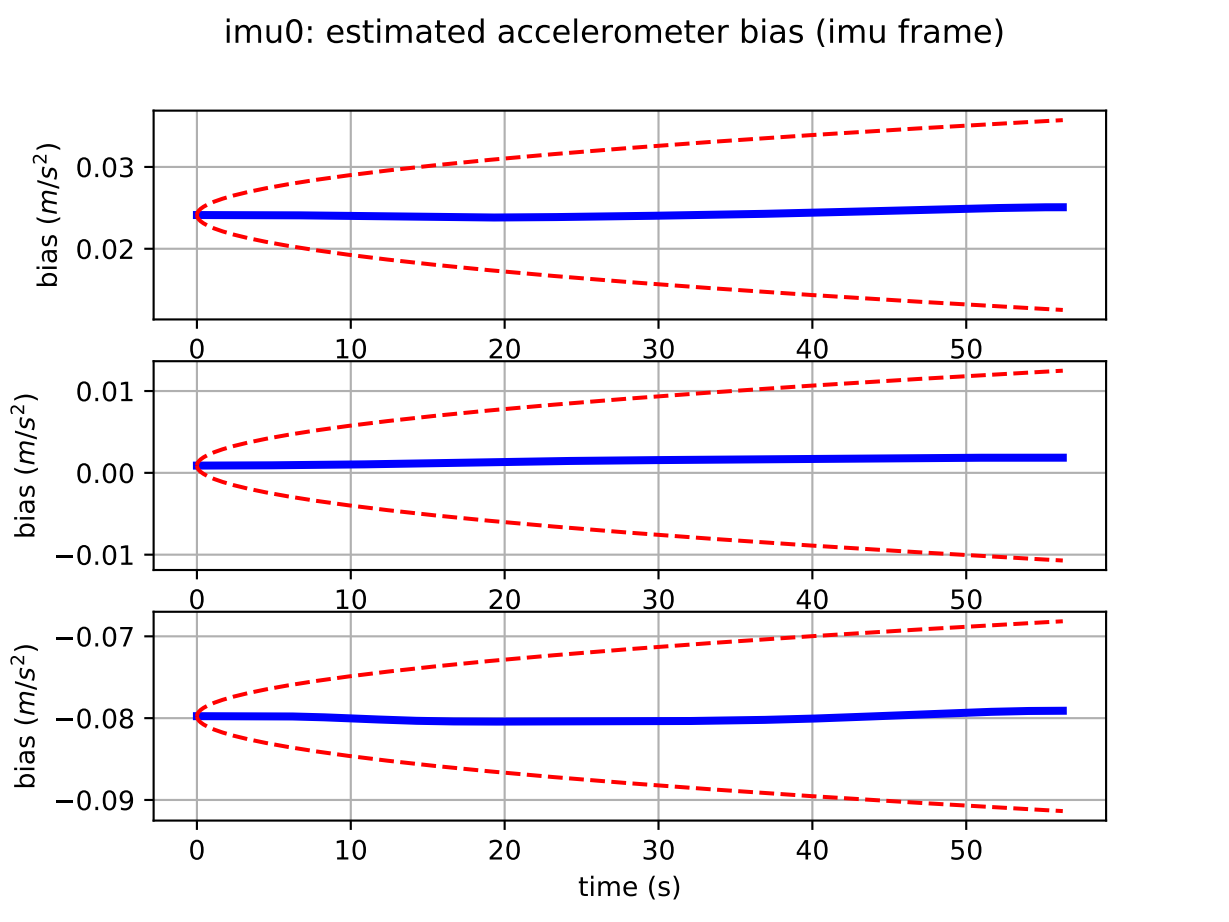

imu0: estimated poses



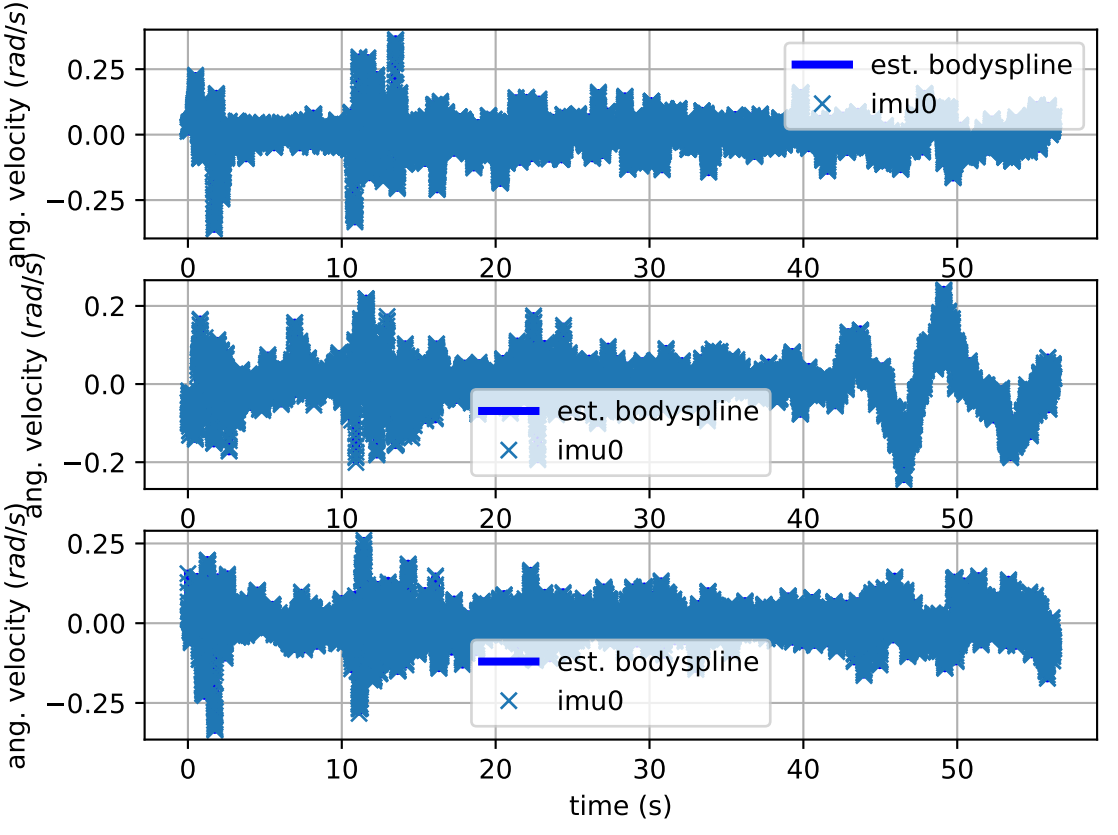

imu0: sample inertial rate
avg dt (ms) = 4.99 +- 0.0065
sample rate (ms)
time (s)
0
1
2
3
4
5
10
20
30
40
50

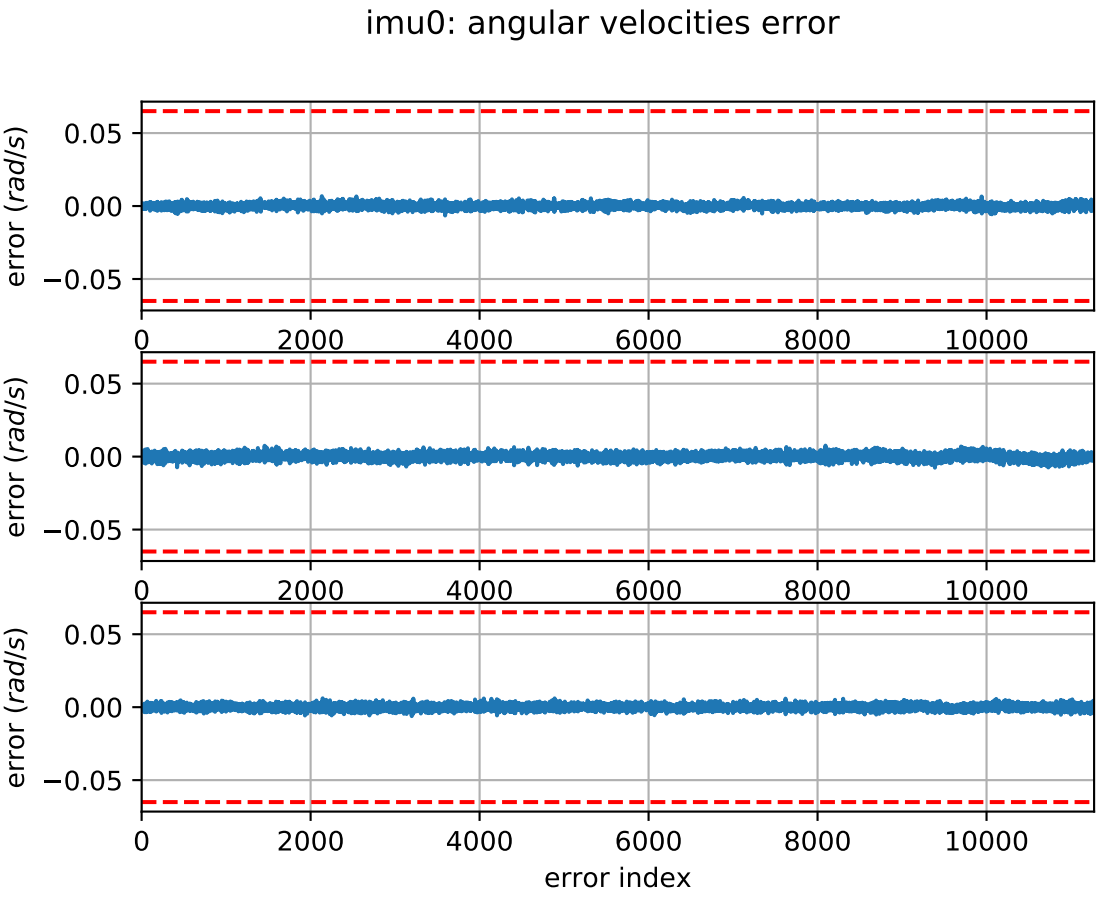

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
1
0
est. bodyspline
imu0
0
10
20
30
40
50
specific force (m/s^2)
−9
−10
est. bodyspline
imu0
0
10
20
30
40
50
specific force (m/s^2)
2
0
est. bodyspline
imu0
0
10
20
30
40
50
time (s)

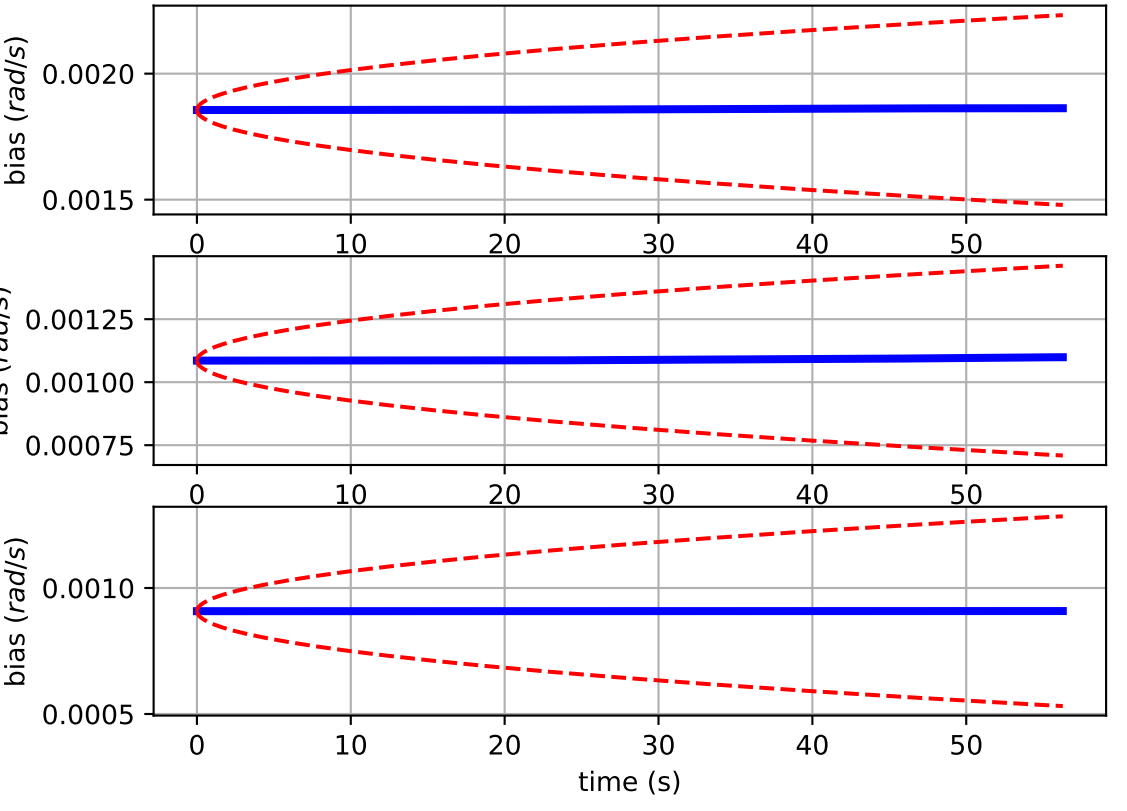

imu0: acceleration error
error (m/s^2)
0.5
0.0
−0.5
0
2000
4000
6000
8000
10000
error index

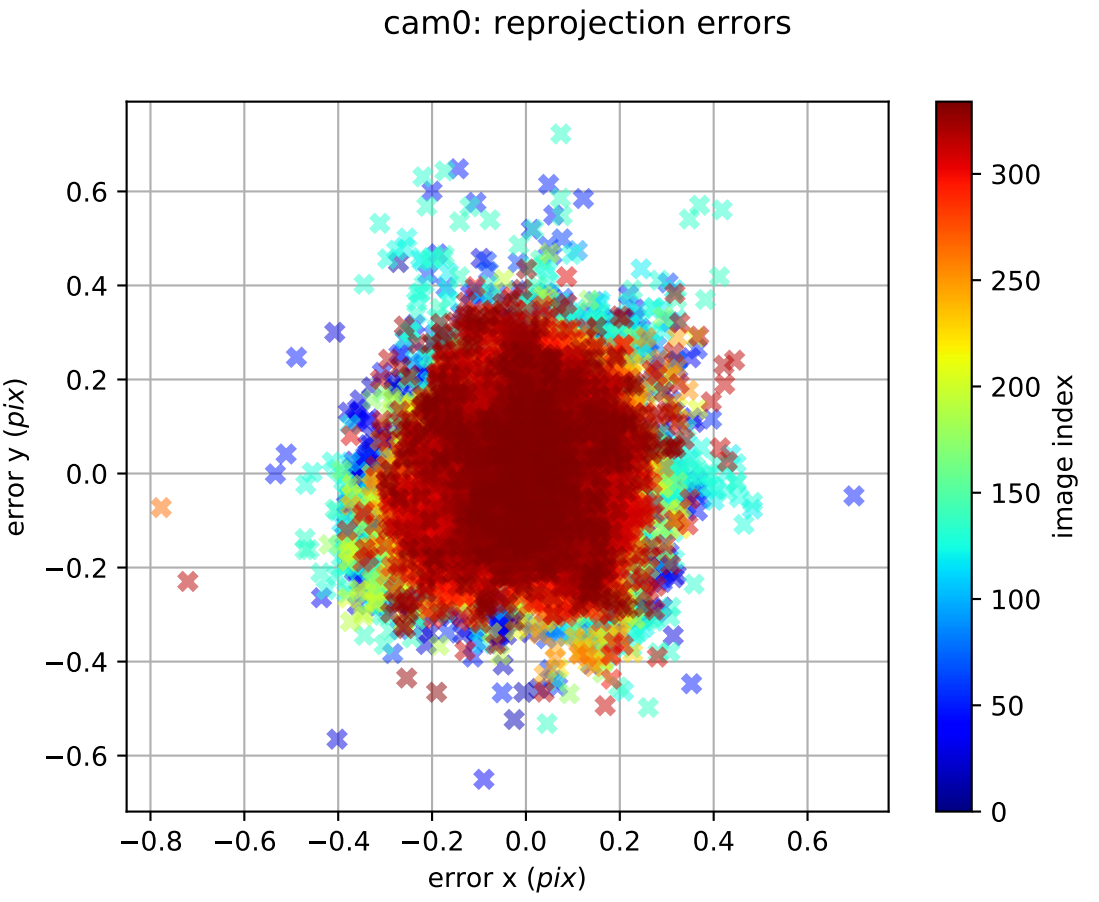

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.03
0.02
0
10
20
30
40
50
bias (m/s^2)
0.01
0.00
−0.01
0
10
20
30
40
50
bias (m/s^2)
−0.07
−0.08
−0.09
0
10
20
30
40
50
time (s)

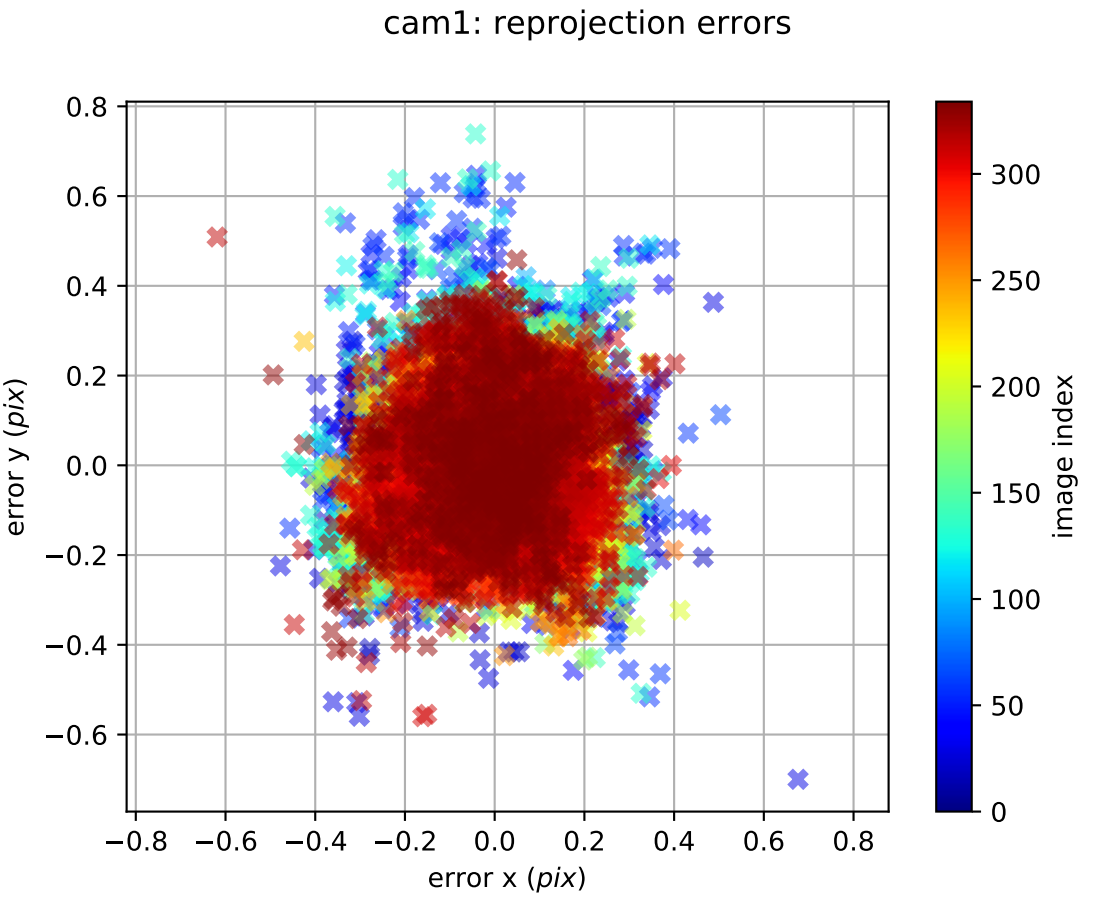
Comparison of predicted and measured angular velocities (body frame)




imu0: angular velocities error
error (rad/s)
0.05
0.00
−0.05
0
2000
4000
6000
8000
10000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.0020
0.0015
0
10
20
30
40
50
0.00125
0.00100
0.00075
0.0010
0.0005
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
300
250
200
150
100
50
0


cam1: reprojection errors
error y (pix)
error x (pix)
image index
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
0
50
100
150
200
250
300